

# Chapter 4, Section 5

James Lee

September 11, 2026

1. Show that a hyperelliptic curve can never be a complete intersection in any projective space.

*Proof.* Combine (Ex. 3.3) and (5.2). □

2. If  $X$  is a curve of genus  $\geq 2$  over a field of characteristic 0, show that the group  $\text{Aut } X$  of automorphisms of  $X$  is finite.

*Proof.* X is hyperelliptic. Let  $\pi : X \rightarrow \mathbb{P}^1$  be the unique  $g_2^1$  of  $X$ . By the Riemann-Hurwitz formula,  $\pi$  is ramified at  $2g + 2$  points. If  $\sigma$  is an automorphism of  $X$ , then there exists  $\tau \in \text{PGL}(1)$  such that the following diagram commutes

$$\begin{array}{ccc} X & \xrightarrow{\sigma} & X \\ \downarrow & & \downarrow \\ \mathbb{P}^1 & \xrightarrow{\tau} & \mathbb{P}^1 \end{array} .$$

This means  $\text{Aut}(X)$  permutes the branch points of  $\pi$ , so there is a group homomorphism  $\text{Aut}(X) \rightarrow S_{2g+2}$ . The latter is a finite group, so it remains to show it has finite kernel. If  $\sigma$  fixes every branch point, then  $\tau$  is the identity, so  $\sigma$  is either the identity map, or the unique hyperelliptic involution (think in terms of function fields and the fact that the hyperelliptic involution is induced by the unique non-trivial element in  $\text{Gal}(K(X)/K(\mathbb{P}^1))$ ).

X is nonhyperelliptic. Consider the canonical embedding  $X \hookrightarrow |K| \cong \mathbb{P}^{g-1}$ . If  $\sigma$  is an automorphism of  $X$ , then there exists  $\tau \in \text{PGL}(g-1)$  such that we have the commutative diagram

$$\begin{array}{ccc} X & \xrightarrow{\sigma} & X \\ \downarrow & & \downarrow \\ \mathbb{P}^{g-1} & \xrightarrow{\tau} & \mathbb{P}^{g-1} \end{array} .$$

By (Ex. 4.6),  $X$  has  $g(g^2 - 1)$  hyperosculation points in this embedding, and  $\tau$  must permute them because linear transformations preserve incidence structures. So there is a group homomorphism  $\text{Aut}(X) \rightarrow S_{g(g^2-1)}$ , and since  $S_{g(g^2-1)}$  is finite, it remains to show the kernel is finite. For every  $g \geq 3$ ,  $X \cap H$  can contain at most point two hyperosculation points for any hyperplane  $H \subset \mathbb{P}^{g-1}$ , i.e., no three hyperosculation points are collinear. It is a general fact that any automorphism in  $\text{PGL}(n)$  is the identity map if it fixes at least  $n + 2$  points in general position. Since  $g(g^2 - 1) \geq g + 1$  for every  $g \geq 3$ , if  $\tau$  fixes the hyperosculation points, it must be the identity. Hence,  $\sigma$  is the identity. As a corollary, any automorphism of a nonhyperelliptic is induced by an automorphism of the canonical embedding, i.e., there exists an injective map  $\text{Aut}(X) \rightarrow \text{PGL}(g-1)$ . □

3. *Moduli of Curves of Genus 4.* The hyperelliptic curves of genus 4 form an irreducible family of dimension 7. The nonhyperelliptic ones form an irreducible family of dimension 9. The subset of those having only one  $g_3^1$  is an irreducible family of dimension 8.

*Proof.* Hyperelliptic elliptic curves of genus 4 are determined as two-fold coverings of  $\mathbb{P}^1$ , ramified at 0, 1,  $\infty$ , and  $2g - 1 = 7$  additional points, up to the action of a certain finite group. So the hyperelliptic curves form an irreducible subvariety of dimension 7 of  $\mathcal{M}_4$  (5.5.5).

The canonical model of a nonhyperelliptic curve of genus 4 is the complete intersection of a unique quadric surface and a cubic surface in  $\mathbb{P}^3$  (5.2.2). The family of all these curves is parametrized by the space  $\mathfrak{S}$  of all complete intersections of a quadric surface and a cubic surface. We can think of  $\mathfrak{S}$  as the set of all homogenous ideals generated by a degree two and three homogenous polynomial in  $k[x_1, \dots, x_4]$ . So there is a morphism  $\mathfrak{S} \rightarrow \mathcal{M}_4$ ,

whose fibres are images of the group  $\mathrm{PGL}(3)$ , which has dimension 15. It remains to compute the dimension of  $\mathfrak{S}$ . There is a projection map  $\mathfrak{S} \rightarrow \Omega$ , where  $\Omega$  is the space of all irreducible quadric surfaces in  $\mathbb{P}^3$ . It is a locally closed subset of  $|\mathrm{H}^0(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(2))|$ , which has dimension  $\binom{3+2}{2} - 1 = 9$ . The fibres of  $\mathfrak{S} \rightarrow \Omega$  is the image of pullback of the linear system of cubic surfaces in  $\mathbb{P}^3$ , which is just  $|\mathrm{H}^0(Q, \mathcal{O}_Q(3))|$  for some quadric surface  $Q \subset \mathbb{P}^2$ . Note that the dimension of this space is independent of the choice of  $Q$ . Consider the short exact sequence

$$0 \longrightarrow \mathcal{O}_{\mathbb{P}^3}(-2) \longrightarrow \mathcal{O}_{\mathbb{P}^3} \longrightarrow \mathcal{O}_Q \longrightarrow 0.$$

Tensoring by  $\mathcal{O}_{\mathbb{P}^3}(3)$  and taking cohomology gives

$$0 \longrightarrow \mathrm{H}^0(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(1)) \longrightarrow \mathrm{H}^0(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(3)) \longrightarrow \mathrm{H}^0(Q, \mathcal{O}_Q(3)) \longrightarrow 0.$$

By exactness, we have

$$\dim |\mathrm{H}^0(Q, \mathcal{O}_Q(3))| = \binom{3+3}{3} - \binom{3+1}{1} - 1 = 15,$$

which implies the fibres of  $\mathfrak{S} \rightarrow \Omega$  have dimension 15, so  $\mathfrak{S}$  has dimension  $9 + 15 = 24$ . Hence,  $\mathfrak{M}_4$  has dimension  $24 - 15 = 9$ . The subspace of  $\Omega$  consisting of quadric cones have dimension 8, so the space of these curves have codimension one in  $\mathfrak{M}_4$ . Hence, the subset of nonhyperelliptic curves having only one  $g_3^1$  is an irreducible family of dimension 8.  $\square$

##### 5. Curves of Genus 5. Assume $X$ is not hyperelliptic.

- (a) The curves of genus 5 whose canonical model in  $\mathbb{P}^4$  is a complete intersection  $F_2.F_2.F_2$  form a family of dimension 12.
- (b)  $X$  has a  $g_3^1$  if and only if it can be represented as a plane quintic with one node. These form an irreducible family of dimension 11.
- (c) In that case, the conics through the node cut out the canonical system (not counting the fixed points at the node). Mapping  $\mathbb{P}^2 \rightarrow \mathbb{P}^4$  by this linear system of conics, show that the canonical curve  $X$  is contained in a cubic surface  $V \subseteq \mathbb{P}^4$ , with  $V$  isomorphic to  $\mathbb{P}^2$  with one point blown up (II, Ex. 7.7). Furthermore,  $V$  is the union of all the trisecants of  $X$  corresponding to the  $g_3^1$  (5.5.3), so  $V$  is contained in the intersection of all the quadric hypersurfaces containing  $X$ . Thus  $V$  and  $g_3^1$  are unique.

*Proof.*

- (a) A curve of genus 5 whose canonical model in  $\mathbb{P}^4$  is a complete intersection of three quadrics is given by a three-dimensional subspace of  $\mathrm{H}^0(\mathbb{P}^4, \mathcal{O}_{\mathbb{P}^4}(2))$ , which has dimension  $\binom{4+2}{2} = 15$ . Thus, the family of all these curves is parametrized by an open subset of the Grassmannian  $\mathrm{Gr}(3, 15)$ , which has dimension  $3(15 - 3) = 36$ . There is a natural action of  $\mathrm{PGL}(4)$  on  $\mathrm{Gr}(3, 15)$ , and since the embedding is canonical, two three-dimensional subspaces in  $\mathrm{Gr}(3, 15)$  give isomorphic curves if and only if they are in the same orbit of  $\mathrm{PGL}(4)$ . So there is a morphism  $\mathrm{Gr}(3, 15) \dashrightarrow \mathfrak{M}_5$ , whose fibres are the images of the group  $\mathrm{PGL}(4)$  which has dimension  $(4 + 1)^2 - 1 = 24$ . Since any individual curve has only finitely many automorphisms (Ex. 5.2), the fibres have dimension = 24, and so the image of  $\mathrm{Gr}(3, 15)$  has dimension  $36 - 24 = 12$ .
- (b) A canonical curve  $X$  of genus 5 in  $\mathbb{P}^4$  has a  $g_3^1$  if and only if it has a trisecant. This means there exists distinct points  $P, Q, R \in C$  such that  $\dim |K - P - Q - R| = 2$  (5.5.3). The embedding of  $X$  in  $|K - P - Q - R| \cong \mathbb{P}^2$  gives a plane quintic with a singular point of multiplicity 2, which is either a cusp or a node. The embedding factors through the canonical embedding followed by a projection map onto a plane in  $\mathbb{P}^4$  away from a trisecant. The proof of (3.10) readily shows there exists an open dense subset of planes in the space of planes in  $\mathbb{P}^4$  which the projection of  $X$  onto the plane gives a curve with only a node.

Conversely, suppose  $X$  can be represented as a plane quintic  $X' \subseteq \mathbb{P}^2$  with one node at  $P \in X'$ . Let  $\pi : V \rightarrow \mathbb{P}^2$  be the blowing up of  $\mathbb{P}^2$  with center at  $P$ , and let  $E \subset V$  be the exceptional divisor of  $\pi$ . Let  $H$  be any hyperplane in  $\mathbb{P}^2$  so that  $|2H|$  is the linear system of conics in  $\mathbb{P}^2$ . Using the results of (V, §3) and (V, §4), the linear system  $|2H - P|$  of conics passing through  $P$  has dimension  $5 - 1 = 4$  with an assigned basepoint at  $P$ . By (II, 7.17.3), the rational map  $\mathbb{P}^2 \dashrightarrow |2H - P| \cong \mathbb{P}^4$  extends to a morphism  $V \rightarrow \mathbb{P}^4$ . It is an embedding of  $V$  into  $\mathbb{P}^4$  because it corresponds to the linear system  $|D|$  with  $D = 2\pi^*H - E$  on  $V$ , which is very ample by (V, 4.2.2). The strict transform  $\tilde{X}'$  of  $X'$  is nonsingular, i.e.,  $\tilde{X}' \cong X$ , and by (V, 3.6), it relates to  $X'$  as divisors in  $V$  in the following way

$$\pi^*X' = \tilde{X}' + 2E.$$

Using the fact that  $X' \sim 5H$  in  $\mathbb{P}^2$ , by (V, Ex. 1) the degree of  $\tilde{X}'$  as a curve in  $\mathbb{P}^4$  equals to

$$\tilde{X}' \cdot D = (5\pi^*H - 2E) \cdot (2\pi^*H - E) = 10(\pi^*H)^2 + 2E^2 = 10 - 2 = 8.$$

By the Riemann-Roch theorem, the only degree  $2g - 2$  invertible sheaf with at least  $g$  sections on a curve of genus  $g$  is the canonical sheaf, so this embedding of  $X$  in  $\mathbb{P}^4$  is canonical. Now, the identification of  $V$  with a rational scroll  $V \rightarrow \mathbb{P}^1$  of degree 3 in  $\mathbb{P}^4$  (V, 2.19.1) induces the following correspondence of divisors on  $V$

$$a\pi^*H - bE = (a - b)C_0 + af,$$

where  $C_0$  is a section and  $f$  is any fibre of the ruling  $V \rightarrow \mathbb{P}^1$ . Rewriting  $\tilde{X}'$  in this way gives

$$\tilde{X}' = 3C_0 + 5f.$$

Since  $\tilde{X}' \cdot f = 3$ , a fibre of  $V \rightarrow \mathbb{P}^1$  and  $\tilde{X}'$  intersect at three points. Also,  $V$  is a rational scroll in  $\mathbb{P}^4$ , so any fibre of  $V$  is a line in  $\mathbb{P}^4$ . Hence, any fibre of  $V \rightarrow \mathbb{P}^1$  is a trisecant of  $X$ , so  $X$  has a 1-parameter family of trisecants in  $\mathbb{P}^4$ .

By (3.11.1), the space  $V_{d,r}$  of irreducible plane curves of degree  $d$  and  $r$  nodes is irreducible and has dimension  $\frac{1}{2}d(d+3) - r$ . In our case,  $V_{5,1}$  has dimension 19, and we have just shown  $V_{5,1}$  parametrizes the space of all genus 5 curves with a  $g_3^1$ . So there is a morphism  $V_{5,1} \rightarrow \mathfrak{M}_5$ , whose fibres are images of the group  $\text{PGL}(2)$ , which has dimension 8. Hence, the family of genus 5 curves with a  $g_3^1$  form an irreducible family of dimension  $19 - 8 = 11$ .

(c) The results of this exercise combined with (5.5.3) can be summarized by the commutative diagram

$$\begin{array}{ccccc} & & \mathbb{P}^2 & & \\ & \nearrow \text{plane quintic} & \uparrow \text{blow up} & & \\ X & \longleftrightarrow & V & \hookrightarrow & \mathbb{P}^4 \cong |K| \cong |D| \\ & \searrow g_3^1 & \downarrow \text{ruled surface} & & \\ & & \mathbb{P}^1 & & \end{array}$$

where  $K$  is a canonical divisor of  $X$ , and  $D$  is the divisor of  $V$  from (Ex. 5.5b). Thus, we have three different representations of nonhyperelliptic curves of genus 5 with a  $g_3^1$ .

□

6. Show that a nonsingular plane curve of degree 5 has no  $g_3^1$ . Show that there are nonhyperelliptic curves of genus 6 which cannot be represented as a nonsingular plane quintic curve.

*Proof.* Let  $X \subset \mathbb{P}^2$  be a nonsingular plane curve of degree 5. The Veronese embedding  $\mathbb{P}^2 \xrightarrow{\sim} V \subset \mathbb{P}^5$  realizes  $\mathbb{P}^2$  as a surface of degree 4, and realizes  $X$  as a curve of degree 10 in  $\mathbb{P}^5$ . Since  $g = 6$ , this embedding is the canonical embedding. In particular, the only degree  $2g - 2$  invertible sheaf with at least  $g$  sections on a curve of genus  $g$  is the canonical sheaf. The proof of (5.5.3) applies to any nonhyperelliptic curve. In particular,  $X$  has a  $g_3^1$  if and only if its canonical embedding in  $\mathbb{P}^5$  has a trisecant line. Suppose that  $L \subset \mathbb{P}^5$  is a trisecant line to the canonical image of  $X$ , meeting  $X$  at three points  $P, Q, R$ . Since  $X \subset V$ , the line  $L$  meets the Veronese surface in  $P, Q, R$ . These three points determine the  $g_3^1$  on  $X$ , and hence  $\dim |P + Q + R| = 1$ . On the other hand, three points of the Veronese surface that are collinear in  $\mathbb{P}^5$  must be collinear in  $\mathbb{P}^2$ , because the line through  $P, Q, R$  intersects  $V$  at three points, and  $V$  has degree 2 in  $\mathbb{P}^5$ , so the line must be contained in  $V$ . Thus  $P, Q, R$  are collinear in  $\mathbb{P}^2$ . But there is only one line in  $\mathbb{P}^2$  passing through three distinct collinear points, contradicting the dimension of  $|P + Q + R|$ . Therefore,  $X$  has no trisecant line in its canonical embedding, and consequently  $X$  has no  $g_3^1$ .

For the second part, it is enough to show there exists nonhyperelliptic curves of genus 6 with a  $g_3^1$ . Consider any irreducible nonsingular curve  $X$  associated to a divisor of type  $(3, 4)$  on the nonsingular quadric surface  $Q$ . It has genus  $(3 - 1)(4 - 1) = 6$ , and one of the families of lines on  $Q$  cuts out a  $g_3^1$  on  $X$ . Hence,  $X$  cannot be represented as a nonsingular plane quintic curve. The canonical sheaf of  $Q$  is  $\omega_Q = \mathcal{O}_Q(-2, -2)$ , so by (II, 8.20), the canonical sheaf of  $X$  is  $\omega_X \cong \mathcal{O}_Q(1, 2) \otimes \mathcal{O}_X$ . Since  $\mathcal{O}_Q(1, 2)$  is a very ample invertible sheaf on  $Q$ ,  $\omega_X$  is very ample as well. Hence,  $X$  is not hyperelliptic by (5.2). □